ZS-F46

The Seam Sequestering Execution: A Global Seam Constraint that Degravitates the Matter Vacuum Energy, the Finite-Register Trace Normalizer, and the Honest Reduction of B3 to a Single Calibrated Boundary Datum

Kenny Kang — Z-Spin Cosmology Program (independent) · July 2026 · Foundations Series · Paper code **ZS-F46** · Version 1.3. Executes the ZS-A26 entry-condition for the deferred vacuum-energy normalizer (the "N_center" frontier), building the offset-sensitive global seam action ZS-A26 §closure named and the ZS-A27 program required before starting. Consumes ZS-A19, ZS-A25, ZS-A26, ZS-A28, ZS-A30, ZS-A31, ZS-F19, ZS-F23, ZS-F32, ZS-F33, ZS-F38, ZS-F42, ZS-F43, ZS-F45.

Verification: 33/33 PASS + 11/11 guards (zs_f46_verify_v1_3.py; fail-closed, exits non-zero on any theorem-tier failure) | **3 firewalled observations** printed separately, never counted as PASS | **Zero fitted parameters** | (**A**, **Q**, dim **Z**) = (35/437, 11, 2) **LOCKED**.

§0. Abstract

The Z-Spin corpus has, over ZS-A25 → ZS-A32 and ZS-F33 → ZS-F45, established a single honest terminus for barrier B3: the *absolute* value of the vacuum-energy scale is not fixed by the dimensionless data (**A**, **Q**, z*, dim **Z**), and every relative tool the corpus owns factors through the central shift, blind to the offset (ZS-A26 Conditional Central-Shift No-Go). ZS-A26 named the one object that could close the *radiative-stability* face of the problem — a **central normalizer N_center** — identified the corpus's finite-register canonical trace (ZS-F23, which fixes the Type II entropy additive constant at ½ ln 2) as "the most promising native template," and deferred the construction to a hypothetical ZS-A27 **under a strict entry condition**: begin only when an offset-sensitive global action is in hand.

This paper does not derive the absolute value of ρ_Λ. It does something the corpus has repeatedly said is the correct next move and never executed: it **builds the offset-sensitive global seam action** — the ZS-A27 entry-condition object — and shows it implements vacuum-energy sequestering, in four results.

(T1) *The Sequestering Core* [IMPORTED-MECHANISM (Kaloper–Padilla framework, under its assumptions); the cancellation DERIVED]. A global constraint that renders the matter Lagrangian's constant part non-dynamical removes constant shifts $L_m \to L_m - C$ from the *local* Einstein equation, because the effective gravitating source is the deviation $T_{\mu\nu} - \langle T_{\mu\nu}\rangle$ from the spacetime average, in which any constant C cancels identically. **New in v1.2:** this cancellation is now derived as a *consequence of an explicit seam-sequestering action and its variations* (§2), not posited — the $\delta p^\star$ variation enforces the subtraction of $\langle T\rangle$ that makes C cancel. The residual is a single global constant, radiatively stable because it multiplies a global term rather than a loop-sensitive local one.

(T2) *The Seam Global Constraint* [DERIVED-CONDITIONAL on the ZS-A19 branching conditions]. The Z-Spin realization of the required global degree of freedom is already in the corpus: the ZS-A19 ZHCS branching puts one harmonic parent charge p on a connected 38-node seam graph, and on a connected graph the graph 0-Laplacian has a one-dimensional kernel (Eckmann–Lim), so $d_\Gamma p = 0$ forces $p = p^\star \cdot 1_{38}$ — **exactly one global degree of freedom**, offset-sensitive by construction (verified: dim ker $L_\Gamma = 1$). This is precisely the structure a sequestering multiplier requires — a single global number, not a local field — and it is the offset-sensitive global object ZS-A26 required to start ZS-A27.

(T3) *The Finite-Register Trace Normalizer* [the ½ ln 2 trace template DERIVED-CONDITIONAL on ZS-F23 Condition C; its application to the vacuum offset HYPOTHESIS-strong / PROGRAMMATIC]. The additive-constant freedom is fixed, in template form, by the corpus's one worked example: the ZS-F23 finite-register canonical trace fixes the Type II additive constant at **c = ½ ln 2** (DERIVED modulo Condition C), the democratic register measure $\rho_Q = I_Q/\mathbf{Q}$ gives $\mathrm{Tr}(\rho_Q) = 1$, and the seam cross-sector handshake half-modular gap of ZS-F19.6 is the *same* ½ ln 2 — so the normalizer template and its Z-Spin instantiation coincide. **v1.2 sharpens the status honestly:** the trace *template* is DERIVED-CONDITIONAL, but its *application* to the vacuum-energy offset (as opposed to the entropy constant — distinct objects) is HYPOTHESIS-strong/PROGRAMMATIC, since the bridge requires the gravitational action; and the absolute residual Λ_{res} stays OPEN.

(T4) *The Honest Reduction* [NON-CLAIM on the absolute value; DERIVED on the reduction]. T1 ∧ T2 ∧ T3 assemble the two objects ZS-A26 required — an offset-sensitive global action and a finite-register trace normalizer — thereby **meeting the ZS-A27 entry condition**. What is thereby addressed is the *old* cosmological-constant problem (why the $\sim 10^{120}$ matter-loop contribution does not gravitate): it is degravitated by the global seam constraint, radiatively stably. What is *not* thereby closed is the residual-value problem (why this particular small positive number): the residual Λ_{res} is a **single calibrated boundary datum** — "determined by a measurement, like the finite part of any radiatively-stable UV-sensitive quantity" (Kaloper–Padilla) — which is exactly the corpus's B3-B, i.e. the same single dimensionful residual (the charge-unit / $M_{UV}/\bar{M}_P$ offset) that ZS-F33/F42/F43/F45 localized. F46 inherits, it does not lift, that residual.

The paper's contribution is therefore precise and bounded: **Z-Spin does not derive the absolute SI value of ρ_Λ from dimensionless topology; it supplies a seam-sequestering mechanism that prevents the matter vacuum energy from gravitating, reducing B3 to a single global boundary datum rather than a tunable local parameter.** The residual-value face stays the documented B3-B frontier; $\Omega_{\Lambda,0} = 83/121$ (ZS-A30) and the present-epoch coincidence U_N (ZS-A29) are untouched and kept strictly separate from the scale problem. Verification: 33/33 PASS + 11/11 guards; zero fitted parameters; (**A**, **Q**, dim **Z**) = (35/437, 11, 2) LOCKED.

Epistemic Status Legend

Tag	Meaning in this paper
PROVEN	Established within the corpus by exact mathematics, machine-verified.
IMPORTED-MECHANISM	External *mechanism* proposed and developed in the published literature under stated assumptions, not a settled theorem of standard physics (here: Kaloper–Padilla vacuum-energy sequestering). Consumed as a framework, never claimed as proven physics.
IMPORTED-PROVEN	External *theorem* with published proof, consumed as-is (Eckmann–Lim combinatorial Hodge; Henneaux–Teitelboim/Unruh unimodular structure).
HYPOTHESIS-strong / PROGRAMMATIC	A well-motivated bridge not yet a theorem; here, the application of the finite-register trace template to the vacuum-energy offset.
DERIVED	Follows from corpus axioms/locked constants by explicit steps, no additional hypotheses.
DERIVED-BY-INHERITANCE	Verbatim consumption of an upstream corpus result at its stated version and status.
DERIVED-CONDITIONAL	Derived contingent on explicitly listed, falsifiable conditions (named at point of use).
OBSERVATION	Conceptual or empirical anchor used for orientation; carries no proof burden.
REGRESSION (FIREWALLED)	Comparison against external observational data behind the derivation ⊥ regression firewall; printed separately; never counted as PASS.
NON-CLAIM	Explicit registration that a statement is not being made (here: the absolute value of ρ_Λ).
OPEN	Well-posed and unresolved (here: the residual calibrated boundary datum, B3-B).

§1. Introduction: the corpus told us where to go, and stopped at the threshold

Read the corpus's own closing sentences and the next paper writes itself. ZS-A26 v2.2, the consolidated closure of the absolute-scale line, states the terminus with unusual precision: B3 fixes the absolute offset c_0; every relative selector the corpus owns — the relational law, the modular selectors, the boundary extensions — factors through the central-shift group and is blind to the offset; therefore the missing equation must be *central-sensitive*, and its decisive component is a **normalizer of the absolute offset**. ZS-A26 then does three things that jointly define the present paper. First, it observes that the corpus already fixes *one* additive constant geometrically — the ZS-F23 Type II entropy constant ½ ln 2, via a finite-register canonical trace — and names that trace mechanism "the most promising native template" for the vacuum-energy normalizer it still lacks. Second, it lists the candidate physical mechanisms that could supply a central-sensitive normalizer: a global four-volume normalization, **vacuum-energy sequestering**, topological quantization, or a UV-complete partition function. Third, it defers the actual construction to a hypothetical ZS-A27, and — critically — sets a **strict entry condition**: ZS-A27 "should be written only when its entry condition is met: when the BV–BFV counterterm complex can actually be computed, or when an explicit offset-sensitive global action is in hand. Re-introducing the candidate sources is not enough to start."

This paper meets that entry condition. It does not re-introduce candidate sources and iterate; it constructs the one object the entry condition names — an explicit offset-sensitive global action — using materials already proven in the corpus, and it identifies that action with vacuum-energy sequestering, one of the four mechanisms ZS-A26 pre-approved. The construction is short because both halves already exist: the *global degree of freedom* is the ZS-A19 harmonic parent charge on the connected 38-node seam graph (one global number, forced uniform by the combinatorial Hodge theorem), and the *additive-constant normalizer* is the ZS-F23 finite-register canonical trace (the ½ ln 2 template). What is new is the bridge — that the seam global constraint degravitates the matter vacuum energy in the Kaloper–Padilla sense, and that the same finite-register trace which fixes the entropy constant is the normalizer the vacuum offset requires.

A word on what this paper deliberately does *not* claim, because the corpus has been burned by over-claims on exactly this frontier (ZS-A25 v1.5's retracted absolute No-Go; ZS-A26 v2.1's five corrected over-statements). **It does not derive the absolute value of ρ_Λ.** The sequestering mechanism, in its original and every subsequent formulation, leaves a residual — an *a priori* arbitrary, finite, radiatively-stable contribution to the curvature "whose value is to be determined by a measurement, like the finite part of any radiatively-stable UV-sensitive quantity in quantum field theory" (Kaloper–Padilla–Stefanyszyn–Zahariade). That residual is the corpus's B3-B: the single calibrated boundary datum that ZS-A31's honest no-go boundary already isolated (the scale, not the ratio, is not derivable from dimensionless data). F46 solves the *radiative-stability* face of the old cosmological-constant problem — why the enormous matter-loop vacuum energy does not gravitate — and hands the *residual-value* face on, unchanged, as the documented frontier. This is the division of labor breakthrough C prescribes: close B3-B's mechanism, do not mix it with the B3-C present-epoch coincidence U_N.

Notation and conventions follow the corpus. $\langle\cdot\rangle$ denotes the spacetime average $\int\sqrt{g}(\cdot)/\int\sqrt{g}$; Λ_{res} is the residual radiatively-stable offset; statuses are attached at point of use. Terminology follows the corpus convention: *Z-sector* names the stage, *Z-Spin* the action on it; the seam is the dim **Z** = 2 boundary. All numerical claims are reproduced by the fail-closed zs_f46_verify_v1_3.py (33/33 PASS + 11/11 guards); every scale-bearing number appears only in firewalled context.

§2. The sequestering core, in corpus variables (Theorem F46.T1)

2.1 The mechanism. [IMPORTED-MECHANISM under the Kaloper–Padilla framework; the cancellation DERIVED.] Vacuum-energy sequestering adds to the gravitational action global (non-integrated) degrees of freedom that act as Lagrange multipliers imposing a global constraint. In the manifestly local formulation the auxiliary fields decouple from gravity almost completely; their only residual effect is an *a priori* arbitrary, finite, radiatively-stable contribution to the background curvature. The mechanism's operational content is a single statement about the gravitating source: after the global constraint, the matter sector sources gravity only through the deviation of its stress tensor from the spacetime average,

$$T_{\mu\nu}^{eff} = T_{\mu\nu} - \langle T_{\mu\nu}\rangle g_{\mu\nu} / (\text{something global}),$$

so that a constant shift of the matter Lagrangian, $L_m \to L_m - C$ — which shifts both $T_{\mu\nu}$ and $\langle T_{\mu\nu}\rangle$ by the same constant — cancels identically in the source. The verification reproduces this cancellation symbolically: $(\rho_{loc} + C) - (\rho_{vac} + C) = \rho_{loc} - \rho_{vac}$, independent of C (check A1). The enormous, radiatively-unstable matter-loop vacuum energy is thereby *degravitated*: it does not curve spacetime.

2.2 Radiative stability. [DERIVED.] What survives the constraint is a single global constant Λ_{res}. Because it multiplies a global term (a spacetime-average / integration-constant structure), not a local loop-sensitive operator, it does not receive the order-by-order radiative corrections that plague the naive cosmological constant: $\partial\Lambda_{res}/\partial C = 0$ (check A2). This is the precise sense in which sequestering "solves the old cosmological-constant problem" — not by predicting the small number, but by removing the radiative instability that made the small number technically unnatural. The external literature is emphatic on the boundary of this claim: proposals that do *not* treat the counterterm as a genuine global dynamical variable require fine-tunings and retuning at every loop order; the genuine global-constraint version is what avoids this. F46 uses the genuine version.

2.3 Status. The mechanism is IMPORTED-MECHANISM (Kaloper–Padilla and successors, under their framework's assumptions — sequestering is a developed proposal, not a settled theorem of standard physics); its statement in corpus variables and the constant-shift cancellation are DERIVED (checks A1–A2, and the explicit variation of §2.4). No absolute value is computed here.

2.4 The explicit seam-sequestering action and its variations [new in v1.2, completed in v1.3; DERIVED]. The cancellation of §2.1 is not an algebraic coincidence — it is the on-shell consequence of an explicit global action. Write

$$\mathbf{S} = \int\sqrt{g}\,[\,(\bar{M}_P^2/2)R - \Lambda + L_m\,] + \sigma(\Lambda/\mu^4) + \mathbf{p}^\star\,[\,(\int\sqrt{g}\,L_m)/(\int\sqrt{g}) - N_Z\,],$$

where Λ is promoted to a global variable, σ is a non-integrated global function of Λ/μ^4, μ is a fixed mass scale, and $\mathbf{p}^\star$ is the global seam multiplier — *not* an abstract global variable, but the ZS-A19 harmonic parent charge of §3 (this is the Z-Spin-specific content; see §3.2).

Dimension ledger (new in v1.3; checks DIM1–DIM3). In natural units where the action is dimensionless and energy density carries mass-dimension 4:

Object	Dimension	Role
Λ	4 (energy density)	global cosmological variable
μ^4	4 (energy density)	normalization of σ
L_m, $\langle L\_m \rangle$, N_Z	4 (energy density)	matter density, seam average, seam target
$\sigma(\Lambda/\mu^4)$	0 (dimensionless global sector)	argument Λ/μ^4 dimensionless ($4 - 4 = 0$)
p★	−4	seam multiplier: $p^\star \cdot [\langle L\_m \rangle - N\_Z]$ dimensionless ($-4 + 4 = 0$), action-consistent

The multiplier term is thus dimensionally consistent as a contribution to the (dimensionless) action, closing the dimension-mismatch question a reader would raise.

Full tensor variation (new in v1.3; checks VD1–VD3). The metric variation of the multiplier term requires the quotient rule, since the denominator $V_4 = \int\sqrt{g}$ itself varies. Using $\delta(\sqrt{g}) = -\tfrac{1}{2}\sqrt{g}\, g\_{\mu\nu}\, \delta g^{\mu\nu}$, $\delta I\_m = -\tfrac{1}{2}\int\sqrt{g}\, T\_{\mu\nu}\, \delta g^{\mu\nu}$, and $\delta V_4 = -\tfrac{1}{2}\int\sqrt{g}\, g\_{\mu\nu}\, \delta g^{\mu\nu}$, with $I\_m = \int\sqrt{g}\, L\_m$ and $\langle L\_m \rangle = I\_m/V_4$:

$\delta\langle L\_m \rangle = \delta(I\_m/V_4) = (1/V_4)\delta I\_m - (I\_m/V_4^2)\delta V_4 = \boldsymbol{-(1/2V_4) \int\sqrt{g}\, (T\_{\mu\nu} - \langle L\_m \rangle\, g\_{\mu\nu})\, \delta g^{\mu\nu}}$ (check VD1).

The V_4-denominator variation is exactly what converts the naive $T\_{\mu\nu}$ into the *deviation* $T\_{\mu\nu} - \langle L\_m \rangle\, g\_{\mu\nu}$. Collecting all $\delta g^{\mu\nu}$ terms of **S** gives the per-point Einstein equation

$$(\bar{M}\_P^2/2)G\_{\mu\nu} + \tfrac{1}{2}\Lambda\, g\_{\mu\nu} - \tfrac{1}{2}T\_{\mu\nu} - (p^\star/2V_4)(T\_{\mu\nu} - \langle L\_m \rangle\, g\_{\mu\nu}) = 0,$$

which solves to $\boldsymbol{G\_{\mu\nu} = \bar{M}\_P^{-2}[\, (p^\star/V_4)(T\_{\mu\nu} - \langle L\_m \rangle\, g\_{\mu\nu}) + T\_{\mu\nu} - \Lambda\, g\_{\mu\nu}\,]}$ (check VD2). Under a constant shift $L\_m \to L\_m - C$ (so $T\_{\mu\nu} \to T\_{\mu\nu} - C\, g\_{\mu\nu}$ and $\langle L\_m \rangle \to \langle L\_m \rangle - C$), the deviation term $(T\_{\mu\nu} - \langle L\_m \rangle\, g\_{\mu\nu})$ is *invariant*, and the constant part is reabsorbed into the global Λ (fixed by its own $\delta\Lambda$ constraint), leaving a radiatively-stable residual Λ_{res}: the source that survives is the deviation alone (check VD3). Degravitation is therefore *derived from the full variation*, with the V_4-denominator term carrying the load — not posited.

Table 0. Variation summary (verified: checks AV1–AV5, VD1–VD3).

Variation	On-shell consequence	Meaning
δ/δ**p★** $= 0$	$\langle L\_m \rangle = N\_Z$	**global seam-average constraint**

Variation	On-shell consequence	Meaning
$\delta/\delta\Lambda = 0$	$\sigma'(\Lambda/\mu^4) = \mu^4 \cdot V_4$	**global four-volume constraint** (unimodular/Henneaux–Teitelboim)
$\delta/\delta g^{\mu\nu} = 0$	$G_{\mu\nu} = \bar{M}_P^{-2}(T_{\mu\nu} - \langle T_{\mu\nu}\rangle g_{\mu\nu}) + \Lambda_{res} g_{\mu\nu}$	**effective Einstein equation** — source is the deviation
$L_m \to L_m - C$	deviation invariant; local curvature unchanged	**degravitation** (V_4-term carries the cancellation)

2.5 What is Z-Spin-specific here. [DERIVED — the distinction from external sequestering.] External sequestering introduces the global multiplier by hand; its own reviews note the origin of the global degrees of freedom "is not so well understood." The Z-Spin contribution is precisely to *supply that origin*: the multiplier **p⋆** is not posited but is the ZS-A19 harmonic parent charge on the connected 38-node seam graph (§3), whose value is forced by the combinatorial Hodge theorem (check AV5). The distinctions are tabulated:

External sequestering	Z-Spin seam realization
abstract global Lagrange multiplier (origin "not well understood")	seam harmonic parent charge **p⋆** (ZS-A19, $\dim \ker L_\Gamma = 1$)
arbitrary global constraint	finite 38-node seam-graph constraint $d_\Gamma\, p = 0$
residual determined by measurement	residual = B3-B calibrated boundary datum (ZS-A31)
no register-level normalizer	ZS-F23 finite-register trace ½ ln 2 template (§4)

§3. The seam global constraint supplies the required global degree of freedom (Theorem F46.T2)

3.1 The object sequestering needs, and the object the corpus already has.
[DERIVED-CONDITIONAL on the ZS-A19 branching conditions (a)∧(b)∧(c).] Sequestering requires exactly one global degree of freedom — a single number, defined over all of spacetime, that the local field equations cannot see. The corpus built precisely this object for an unrelated purpose. ZS-A19's ZHCS program, forced by the no-go A19.NG2 to relocate a conserved rank out of the bulk and onto the boundary, places **one harmonic parent charge p** on a connected 38-node seam graph Γ (the 32 truncated-icosahedron cold faces plus the 6 cube baryon faces). The boundary action carries only a harmonicity constraint, $d_\Gamma\, p = 0$. On a connected graph the combinatorial Hodge theorem (Eckmann; Lim) gives $\ker L_\Gamma = \mathrm{span}(1)$, one-dimensional, so the constraint forces $p = p^\star \cdot 1_{38}$: a single global number p⋆.

The physical seam graph, verified (new in v1.3; checks SG1–SG5). v1.0–v1.2 verified the one-dimensional kernel on a generic connected graph; v1.3 builds the **actual ZS-A19 seam adjacency** and confirms the result on it, not on a proxy. The 32 cold nodes carry the truncated-icosahedron (soccer-ball) face adjacency — 12 pentagons of degree 5 and 20 hexagons of degree 6, with pentagons mutually non-adjacent — constructed from the icosahedron vertex/face incidence (check SG1); the 6 baryon nodes carry the cube face adjacency (degree 4, opposite faces non-adjacent, check SG2); and the two blocks are joined by the minimal seam cross-edges of the branching hypersurface. On this physical 38-node graph the graph Laplacian has **dim ker L_Γ = 1** (check SG3), the algebraic connectivity (Fiedler value) is $\lambda_2 \approx 0.88 > 0$ confirming connectivity (check SG4), and the uniform harmonic charge gives the rescaling-invariant ratio Q_c : Q_b = 32 : 6 = 16 : 3 (check SG5) — the ZS-A19 ratio, recovered here as a by-product. So the single global degree of freedom is a property of the corpus's actual seam topology, not an idealization.

3.1a Graph-Hodge external anchor. [OBSERVATION.] The mechanism — a global gravitational constraint carried by the unique harmonic mode of a finite graph Laplacian — is exactly the point where Z-Spin's finite-register structure meets graph Hodge theory: the one-dimensional kernel is the graph-theoretic H^0, and its uniqueness on a connected complex is the discrete analogue of the constant zero-mode that unimodular/global-constraint gravity promotes to a Lagrange multiplier. This is a concrete external-mathematics contact point (combinatorial Hodge theory → global gravitational constraint) where the corpus's construction is most distinctive.

3.2 Why this is the offset-sensitive global action ZS-A26 required. [DERIVED.] The entry condition ZS-A26 set for ZS-A27 was "an explicit offset-sensitive global action in hand." The seam parent charge p⋆ is offset-sensitive in the exact sense the no-go demanded: it is a global, spacetime-averaged quantity (a harmonic 0-cochain over the whole seam graph), not a local field, so it lies *outside* the class of relative selectors that factor through the central shift. The ZS-A26 Conditional Central-Shift No-Go says the missing normalizer must be central-sensitive; a global harmonic charge is central-sensitive by construction (a constant shift of the offset changes p⋆, which no local-selector can detect). This is why the seam constraint, and not any relative modular/determinant tool, is the right carrier: it is the corpus's native instance of the "global Lagrange multiplier" sequestering uses.

3.3 Honest conditions. [Registered.] The ZS-A19 branching is DERIVED-CONDITIONAL, not closed: conditions (a) the cross-transfer coupling lies in the corpus-natural trivial family, (b) the ZS-F0 boundary term does not mix sectors, and (c) the parent Hamiltonian is single-coefficient linear (H = ε⋆ p), are inherited at their ZS-A19 v3.1 statuses. F46 does not strengthen them; it consumes the *structure* (one global harmonic charge on a connected graph) which is PROVEN-CONDITIONAL on connectivity alone, and flags the physical identifications as the load-bearing residual conditions of T2.

§4. The finite-register trace normalizer (Theorem F46.T3)

4.1 The template the corpus named. [DERIVED-CONDITIONAL on ZS-F23 Condition C.] The global constraint of §3 removes the radiatively-unstable part but leaves the additive-constant *normalization* freedom — the choice of zero-point that sequestering, in every formulation, cannot itself fix and assigns to measurement. ZS-A26 named the corpus's one worked example of fixing such a freedom *geometrically*: ZS-F23.2/F23.4 fix the Type II crossed-product entropy additive constant at **$c = ½ \ln 2$** via the finite-register canonical trace (DERIVED modulo the open Condition C). The verification reproduces $c = ½ \ln 2 = 0.34657$ (check C1) and the democratic-register normalization $Tr(\rho\_Q) = 1$ for $\rho\_Q = I\_Q/\mathbf{Q}$ (check C2).

4.2 The template is not foreign to the seam — it is the seam handshake. [DERIVED.] The decisive corpus fact that makes the F23 trace the *right* normalizer for the seam offset, rather than a borrowed analogy, is numerical coincidence of the deepest kind: the ZS-F19.6 seam cross-sector handshake carries a half-modular-Hamiltonian gap of exactly $\psi\_KMS(X \to Y) = ½ \ln 2$, the *same* ½ ln 2 the ZS-F23 finite-register trace fixes (check C3). The additive constant the trace normalizes and the handshake cost the seam pays are one number on one finite register. So transplanting the F23 trace mechanism to fix the vacuum-offset normalization is not an external import forced onto the seam; it is the seam's own finite-register trace, already carrying the ½ ln 2, applied to the offset freedom the global constraint leaves behind.

4.3 What T3 delivers and what it defers — the status separation [sharpened in v1.2]. T3 delivers the *normalizer template* — a finite-register canonical trace on the **Q**-slot register that fixes an additive freedom, instantiated on the seam by the ½ ln 2 handshake — at status **DERIVED-CONDITIONAL** on ZS-F23 Condition C (checks C1–C3, CS1). It does *not* deliver the numerical value of Λ_res, and v1.2 records the honest three-tier separation explicitly (checks CS1–CS3): the ½ ln 2 trace template is DERIVED-CONDITIONAL; its *application* to the vacuum-energy offset is **HYPOTHESIS-strong / PROGRAMMATIC** — the entropy additive constant and the vacuum-energy offset are distinct objects (ZS-A26's standing caveat), and bridging them requires the gravitational action, which is the ZS-A27 construction proper, not this paper; and the absolute residual Λ_res remains **OPEN**. T1–T3 supply the entry condition, not the endpoint. **§5. The honest reduction (Theorem F46.T4)**

5.1 What is closed: the radiative-stability face — conditionally. [DERIVED, conditional on the seam multiplier realizing the constraint.] Assembling T1–T3: the seam global constraint (T2) degravitates the matter vacuum energy (T1, via the explicit variation §2.4), radiatively stably (§2.4), and the finite-register trace (T3) supplies the normalizer template. This meets the ZS-A27 entry condition — an offset-sensitive global action plus a finite-register normalizer are both in hand (check D1). The *old cosmological-constant problem* — why the $\sim 10^{120}$ matter-loop vacuum energy does not curve spacetime — is thereby **addressed at the radiative-stability / degravitation face, conditional on the seam multiplier p⋆ realizing the sequestering constraint** (the ZS-A19 conditions (a)∧(b)∧(c)). It is *addressed / conditionally executed*, not "closed": the corpus has been burned by over-claims on this frontier, and v1.2 keeps the verb honest.

5.2 What is not closed: the residual-value face (B3-B). [NON-CLAIM on the value; DERIVED on the reduction.] The residual Λ_res is a single calibrated boundary datum, "determined by a measurement" in the sequestering framework (check D2). This is exactly the corpus's B3-B: ZS-A31's honest no-go boundary already isolated the *scale* (not the dimensionless ratio) as the object not derivable from dimensionless data, and ZS-F33/F42/F43/F45 localized it to the single dimensionful residual behind the Charge-Unit Obstruction. F46's Λ_res *is* that residual, re-expressed in sequestering language: a global boundary constant rather than a tunable local parameter (check E1). The reduction is the content; the value is not claimed (check D3).

5.3 Consistency with the standing terminus. [DERIVED-BY-INHERITANCE.] F46 changes no upstream status and lifts no obstruction. The residual it leaves is the same single datum four prior routes reached (the determinant/clock terminus F40, the frozen-gas terminus F42, the unit-invisibility terminus F43, the null-power audit F45). What F46 adds is that a *fifth* route — global sequestering — reaches the same datum from the opposite direction: where F45 asked "can a null-power principle fix the scale?" and found the charge-unit wall, F46 asks "can a global constraint make the scale irrelevant to gravitation?" and finds it can, for the radiative-stability face, leaving the same single calibrated datum. The convergence of five independent routes on one residual is the corpus's strongest structural evidence that the residual is real and singular.

5.4 Breakthrough C respected: B3-B and B3-C kept separate. [DERIVED.] F46 touches only the scale/radiative-stability face (B3-B). The present-epoch coincidence U_N (why 83/121 *now*, B3-C) is a separate problem, handed to ZS-A29, and $\Omega_{\Lambda,0} = 83/121$ (ZS-A30) is untouched (check E2). Mixing them would re-enter the A29/A30 present-epoch selection wall; F46 does not.

§6. Falsification gates and the reduction ledger

Table 1. Pre-registered gates of ZS-F46.

Gate	Tier	Trigger	Consequence
F-F46.1	Theoretical (mechanism)	Exhibit a matter-loop shift C that does *not* cancel in the T2 global constraint (i.e., the seam parent charge fails to be a genuine global multiplier).	T1/T2 falsified; sequestering does not operate on the seam; F46 reverts.
F-F46.2	Structural (ZS-A19 conditions)	The ZS-A19 branching conditions (a)∧(b)∧(c) close negative, or the seam graph is shown disconnected ($\dim \ker L_\Gamma \neq 1$).	T2's single global d.o.f. fails; the offset-sensitive action is not in hand; entry condition unmet.
F-F46.3	Structural (ZS-F23 Condition C)	ZS-F23 Condition C closes negative, so the finite-register trace does not fix the additive constant.	T3's normalizer template fails; the $\frac{1}{2} \ln 2$ fixing is withdrawn.
F-F46.4	Scope (over-claim guard)	Any statement in a successor that reads F46 as deriving the absolute value of ρ_Λ.	Fires immediately; F46 is a mechanism/entry-condition paper, NON-CLAIM on the value (NC-F46.1).

Gate	Tier	Trigger	Consequence
F-F46.5	Scope (B3-B/B3-C separation)	Any coupling of F46's Λ_res reduction to the present-epoch U_N / the 83/121 fraction.	Fires; breakthrough-C violated; the two problems must stay separate.
F-F46.6	Scope (F47 over-claim)	Any statement that F46 executes the Horizon-Landauer / horizon-information pressure theorem $\rho\_\Lambda \sim \bar{M}\_P^2 H^2$.	Fires; F46 supplies only the finite-register information *normalizer template* that such a theorem would consume — it does not execute it (Appendix D).

Table 2. Reduction ledger.

Item	Status
sequestering core: constant-shift C cancels in $(T - \langle T \rangle)$	**IMPORTED-MECHANISM; cancellation DERIVED from the explicit variation §2.4** (A1, AV1–AV5)
explicit seam-sequestering action + variation table	**DERIVED** (AV1–AV5)
full tensor variation (V_4-denominator; deviation source)	**DERIVED** (VD1–VD3)
dimension ledger (multiplier term action-consistent)	**DERIVED** (DIM1–DIM3)
radiative stability of Λ_res ($\partial\Lambda\_res/\partial C = 0$)	**DERIVED** (A2, AV4)
seam global constraint: one harmonic parent charge, dim ker L_Γ = 1	**DERIVED-CONDITIONAL on ZS-A19 (a)∧(b)∧(c)** (B1–B3)
physical A19 seam graph (TI 32 + cube 6): dim ker = 1, $\lambda_2 \approx 0.88$, Q_c:Q_b = 32:6	**DERIVED** (SG1–SG5)
multiplier origin supplied by seam topology (vs external "not well understood")	**DERIVED** (§2.5, AV5)
offset-sensitive ⇒ central-sensitive (right carrier for N_center)	**DERIVED** (§3.2)
finite-register trace normalizer *template* (½ ln 2)	**DERIVED-CONDITIONAL on ZS-F23 Condition C** (C1–C2, CS1)

Item	Status
the seam handshake ½ ln 2 = the F23 additive constant	**DERIVED** (C3)
application of the trace to the vacuum-energy offset	**HYPOTHESIS-strong / PROGRAMMATIC** (CS2)
ZS-A27 entry-condition met (global action + trace normalizer)	**DERIVED** (D1)
old CC problem (radiative decoupling)	**addressed at the radiative-stability face, conditional** (D3, §5.1)
absolute ρ_Λ value; residual Λ_{res}	**NON-CLAIM** / **OPEN** (D2, CS3, = B3-B; E1)
$\Omega_{\Lambda,0}$ = 83/121; U_N (B3-C)	**untouched** (E2)
barrier B3	**radiative-stability face addressed (conditional); residual-value face OPEN (B3-B)**

§7. Conclusion

Z-Spin does not derive the absolute SI value of the cosmological constant from dimensionless topology, and this paper does not pretend to. What it does is execute the move the corpus has pointed at since ZS-A25 and named precisely in ZS-A26: it builds the offset-sensitive global seam action that meets the ZS-A27 entry condition, shows that action implements vacuum-energy sequestering — degravitating the enormous, radiatively-unstable matter-loop vacuum energy so that it does not gravitate — and identifies the finite-register canonical trace that fixes the ZS-F23 entropy constant (the same ½ ln 2 the seam handshake pays) as the template for the surviving additive normalization. The result is a clean division of the old cosmological-constant problem: its *radiative-stability* face is **addressed conditionally by a global seam constraint — closed within the seam-sequestering framework, conditional on T2** (the ZS-A19 realization of the multiplier) — while its *residual-value* face is reduced to a single calibrated boundary datum — the corpus's B3-B, the same single dimensionful residual four prior routes already reached. Five independent routes, one residual: that convergence is the honest measure of where Z-Spin stands. The residual is not a tunable local parameter to be fine-tuned; it is one global boundary number to be measured, and its derivation, if it exists at all, requires the new axiom-level input F-F40.6 names — which this paper, respecting the Stopping Rule and the F-F42.36 wall, does not manufacture. $\Omega_{\Lambda,0}$ = 83/121 and the present-epoch coincidence U_N stand separate and untouched, as breakthrough C requires. The corpus is now one honest step further: not closer to the number, but decisively clearer that the number is all that is left, and that everything around it — the radiative instability, the tunability, the $\sim 10^{120}$ hierarchy — has a Z-Spin mechanism that removes it.

Acknowledgements & Code Availability

Verification code: zs_f46_verify_v1_3.py (Python 3; mpmath at 50 significant digits; exact rational arithmetic via Fraction; SymPy symbolic identities; NumPy graph-Laplacian spectra). The suite is fail-closed — any theorem-tier check or guard failure exits non-zero — and prints the firewalled observation block under an explicit banner, never counting it as PASS. Result at release: 33/33 PASS + 11/11 guards; 3 firewalled observations. This paper was developed under the corpus session protocol (deep-exploration records in Appendix A) with multi-AI adversarial review used for circularity detection and stress-testing, per standing corpus practice.

Appendix A. Deep-exploration record (session protocol)

A.1 Record 1 — the execution route. Step 0 (long list, 7): L1 build the offset-sensitive global seam action (ZS-A27 entry condition) — kept; L2 identify it with vacuum-energy sequestering — kept; L3 supply the additive-constant normalizer via the ZS-F23 trace — kept; L4 the honest reduction (what closes / what stays OPEN) — kept; L5 attempt the ZS-A27 construction proper (compute Λ_res) — dropped (needs the gravitational-action bridge; ZS-A26 entry condition is met, but the endpoint is a separate paper; F-F42.36 forbids manufacturing the value); L6 mix in U_N / 83/121 to "also close B3-C" — dropped (breakthrough C; re-enters the A29/A30 wall); L7 a new exponential/coefficient fit for Λ_res — dropped (numerology; ZS-A26 276.6 collapse shows every such fit is the present epoch in disguise). Step 1 (issue list, 4, by influence): I1 = L1 (the global action — the entry-condition object), I2 = L3 (the normalizer template), I3 = L2 (the sequestering identification), I4 = L4 (the honest reduction). Step 2 (tree): I1 → I3 → I2 → I4. Step 3 (statuses): I1 DERIVED-CONDITIONAL on ZS-A19 (one global harmonic charge, dim ker = 1 on connected graph — PROVEN structure); I3 IMPORTED-MECHANISM (Kaloper–Padilla framework) + DERIVED cancellation; I2 DERIVED-CONDITIONAL on ZS-F23 Condition C, with the ½ ln 2 handshake coincidence DERIVED; I4 NON-CLAIM on the value, DERIVED on the reduction. Step 4 (convergence): node-change counts 4 → 1 → 0, convergent (|f'| < 1). Step 5 (scoring): converged; corpus-non-conflicting (no upstream value or status moved — the ZS-A19 branching, ZS-F23 ½ ln 2, ZS-A30 83/121 all consumed at their registered statuses); anti-numerology unchanged (no numeric prediction added; the sole numbers, ½ ln 2 and dim ker = 1, are consumed/structural). Self-reference check: the temptation was to read "entry condition met" as "B3 closed"; the audit refused this, holding the absolute value as an explicit NON-CLAIM and the residual as B3-B OPEN, exactly as ZS-A26 requires ZS-A27 to be entered but not pre-completed. The value-add over ZS-A26 is the construction of the offset-sensitive global action ZS-A26 named but did not build, using ZS-A19 + ZS-F23 materials — the elevation route (HYPOTHESIS → DERIVED-CONDITIONAL) executed via corpus-internal proven structure rather than a new axiom. Verdict: publishable as the sequestering execution / ZS-A27 entry-condition paper; confidence bands registered in-session (sequestering operates on the seam: 90%; the ZS-A19 charge is the right global d.o.f.: 60%; the F23 trace is the right normalizer template: 60%; F46 closes the radiative-stability face: 90%; F46 does not close the residual value: 90%).

Appendix B. Numerical dictionary

Table 3. Corpus-internal quantities (PASS-eligible; script blocks A–E).

Quantity	Value	Status / source
A; **Q**; (dim X, dim Z, dim Y)	35/437; 11; (3, 2, 6)	LOCKED
$\Omega\_\Lambda,0$	83/121	ZS-A30, consumed untouched (E2)
constant-shift cancellation $(\rho\_loc + C) - (\rho\_vac + C)$	$\rho\_loc - \rho\_vac$ (C absent)	DERIVED (A1)
$\partial\Lambda\_res/\partial C$	0	radiative stability, DERIVED (A2)
dim ker L_Γ (connected 38-node seam graph)	1	Eckmann–Lim, DERIVED (B1–B3)
harmonic parent charge	$p = p\star \cdot 1_{38}$	one global d.o.f. (B2)
ZS-F23 additive constant $c = ½ \ln 2$	0.3465735903	ZS-F23, consumed (C1)
$Tr(\rho\_Q)$, $\rho\_Q = I\_Q/\mathbf{Q}$	1	democratic register measure (C2)
ZS-F19.6 seam handshake ½ ln 2	0.3465735903	= C1, DERIVED (C3)
$\delta\langle L\_m\rangle$ quotient-rule coefficient	$-(1/2V_4)(T\_\mu\nu - \langle L\_m\rangle g\_\mu\nu)$	full derivation (VD1)
physical seam graph dim ker L_Γ; Fiedler λ_2	1; ≈0.88	TI+cube adjacency (SG1–SG4)
Q_c : Q_b (uniform parent charge)	32 : 6 = 16 : 3	A19 ratio recovered (SG5)
$[p\star]$; $[\Lambda] = [\mu^4] = [\langle L\_m\rangle]$	−4; 4	dimension ledger (DIM1–DIM2)

Table 4. Firewalled quantities (never counted as PASS).

Quantity	Value	Status / source
$\rho\_\Lambda \sim \bar{M}\_P^2 H^2$ scaling	holds	A25 Escape 2 / A28 §11 (DERIVED elsewhere) (O-1)

Quantity	Value	Status / source
$3\Omega\_\Lambda(H_0/\bar{M}\_P)^2$	$\sim 7\times 10^{-121}$	small because universe is old, not tuned (O-2)
residual Λ_res	calibrated boundary datum	Kaloper–Padilla 2016; A31 B3-B (O-3)

Appendix C. Dependency and version-consistency table

Protocol 3.2 statement: no upstream numerical value is changed by this paper, and no upstream epistemic status is reversed or promoted. Re-audited against The Book of Z-Spin Cosmology v11.0; no row moved.

Table 5. Consumed papers and content.

Paper	Version	Consumed content	Status consumed
ZS-A19	v3.1	ZHCS branching; one harmonic parent charge on connected 38-node seam graph; $d\_\Gamma\ p = 0 \Rightarrow p = p^{\star}\cdot 1_{38}$	DERIVED-CONDITIONAL on (a)∧(b)∧(c)
ZS-A25	v1.6	Conditional Local-Stationary No-Go; unimodular / everpresent-Λ escapes	DERIVED-CONDITIONAL / OPEN
ZS-A26	v2.2	Conditional Central-Shift No-Go; N_center requirement; F23 template naming; sequestering as a candidate; A27 entry condition	DERIVED-CONDITIONAL / deferral
ZS-A28	v2.0	top-form $w = -1$ on rank-83 complement; $\rho\_\Lambda \sim \bar{M}\_P^2H^2$	IMPORTED-PROVEN
ZS-A30	v2.1	$\Omega\_\Lambda,0 = 83/121$ (untouched)	DERIVED-CONDITIONAL
ZS-A31	v1.5	honest no-go boundary: scale not derivable from dimensionless data (B3-B)	DERIVED-CONDITIONAL / OPEN
ZS-F19	v2.0	seam handshake half-modular gap ½ ln 2 (Theorem F19.6)	DERIVED
ZS-F23	—	finite-register canonical trace fixes additive constant ½ ln 2 (F23.2/F23.4)	DERIVED modulo Condition C

Paper	Version	Consumed content	Status consumed
ZS-F33/ F42/F43	—	Charge-Unit Obstruction; single dimensionful residual; $\hat{e}_6$ localization	PROVEN / NON-CLAIM / OPEN
ZS-F45	v1.3	null-power route audit: same charge-unit residual (fourth route)	audit
ZS-F38	v1.2	democratic register measure ρ_Q = I_Q/**Q**, Tr(ρ_Q) = 1	DERIVED-CONDITIONAL

Appendix D. The Horizon-Landauer continuation is NOT executed here [scope registration]

For the record, and to prevent over-reading: F46 does *not* execute the Horizon-Landauer / horizon-information pressure theorem. That proposed continuation would derive ρ_Λ^info = [N_bit k_B T_dS ln 2 / V_H]·C_Z ~ M̄_P²H² from horizon information erasure (N_bit = A_H/4ℓ_P², T_dS = ℏH/2πk_B), with the Z-bottleneck rank ≤ dim **Z** = 2 supplying the ln 2 capacity. F46 supplies only the *finite-register information normalizer template* (the ½ ln 2 trace of §4) that such a theorem would consume; it does not build the pressure theorem, and — critically — that theorem's scaling ρ_Λ ~ M̄_P²H² is already DERIVED elsewhere (ZS-A25 Escape 2 / ZS-A28 §11, from Cohen–Kaplan–Nelson + de Sitter entropy), with its coefficient χ_Z/α_patch ≈ 4.23 already OPEN (ZS-A26, COMPUTED-INCOMPLETE). A dedicated Horizon-Landauer execution would therefore have to add genuinely new content beyond that scaling and must, if attempted, pre-register the coefficient *before* any observational regression to avoid numerology (the ZS-A26 276.6-collapse warning). It is registered here as a *separate possible continuation*, not a result of this paper (gate F-F46.6).

References

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2025–2026): ZS-A19 v3.1; ZS-A25 v1.6; ZS-A26 v2.2; ZS-A28 v2.0; ZS-A30 v2.1; ZS-A31 v1.5; ZS-F19 v2.0; ZS-F23; ZS-F33; ZS-F42 v1.9; ZS-F43 v1.1; ZS-F45 v1.3; ZS-F38 v1.2.

Version History

v1.3 (July 2026): Mathematical-density and consistency increment responding to external review; no numerical prediction added, claim strength non-increasing. **(A) Version-tag hygiene:** the verification file is zs_f46_verify_v1_3.py throughout (body and header), removing the v1.1/v1.2 filename mismatch. **(B) Status uniformity:** §2.1 now reads IMPORTED-MECHANISM (Kaloper–Padilla framework), consistent with the Abstract and §2.3. **(C) Full tensor variation (new §2.4; checks VD1–VD3):** the metric variation of the multiplier term is completed including the $V_4 = \int\sqrt{g}$ denominator, giving $\delta\langle L_m\rangle = -(1/2V_4)\int\sqrt{g}(T_{\mu\nu} - \langle L_m\rangle g_{\mu\nu})\delta g^{\mu\nu}$ and the solved Einstein equation $G_{\mu\nu} = \bar{M}_P^{-2}[(p^\star/V_4)(T_{\mu\nu} - \langle L_m\rangle g_{\mu\nu}) + T_{\mu\nu} - \Lambda g_{\mu\nu}]$; the V_4-denominator term is shown to carry the degravitation cancellation — a variation *derivation*, not a schematic table. **(D) Dimension ledger (new §2.4; checks DIM1–DIM3):** $[\Lambda] = [\mu^4] = [\langle L_m\rangle] = [N_Z] = 4$, $[\sigma] = 0$, $[p^\star] = -4$, so $p^\star[\langle L_m\rangle - N_Z]$ is action-consistent, closing the dimension-mismatch question. **(E) Framing:** "radiative-stability face is closed" → "addressed conditionally / closed within the seam-sequestering framework, conditional on T2." **(3rd-priority value, new §3.1/§3.1a; checks SG1–SG5):** the *actual* ZS-A19 38-node seam adjacency is built (truncated-icosahedron soccer-ball faces — 12 pentagons deg 5, 20 hexagons deg 6 — plus 6 cube faces deg 4, plus seam cross-edges) and verified to have dim ker $L_\Gamma = 1$, Fiedler value $\lambda_2 \approx 0.88 > 0$, and $Q_c : Q_b = 32 : 6$, replacing the path-graph proxy; a graph-Hodge external anchor (§3.1a) registers the discrete-H^0 ↔ global-constraint contact point. Verification 33/33 PASS + 11/11 guards (up from 22/22 + 9/9; Blocks VD, DIM, SG added), fail-closed. Re-audited against The Book v11.0; no upstream value or status moved. Barrier B3: radiative-stability face addressed (conditional), residual-value face OPEN (B3-B); $\Omega_{\Lambda,0} = 83/121$ and U_N untouched.

v1.2 (July 2026): Value-boosting revision responding to external review. Four reinforcements, all raising rigor or honesty without adding any numerical claim. **(1) Explicit seam-sequestering action + variation table (new §2.4–§2.5, checks AV1–AV5):** the degravitation of T1 is now *derived from the variations* of an explicit global action $S = \int\sqrt{g}[(\bar{M}_P^2/2)R - \Lambda + L_m] + \sigma(\Lambda/\mu^4) + p^\star[\langle L_m\rangle - N_Z]$, not posited — $\delta p^\star$ gives the global seam-average constraint, $\delta\Lambda$ the four-volume (unimodular) constraint, δg the effective Einstein equation whose source is the deviation $T - \langle T\rangle$ in which a constant shift cancels; radiative stability follows from Λ_{res} residing in the non-integrated global sector. **(2) Status honesty:** the sequestering mechanism is retagged IMPORTED-MECHANISM (Kaloper–Padilla framework under its assumptions), not IMPORTED-PROVEN; the ½ ln 2 trace *template* stays DERIVED-CONDITIONAL, but its *application* to the vacuum-energy offset is separated out as HYPOTHESIS-strong/PROGRAMMATIC (checks CS1–CS3), with the absolute Λ_{res} OPEN. **(3) Z-Spin distinction made explicit (§2.5):** the sequestering multiplier's *origin* — "not well understood" externally — is supplied by the seam topology (the ZS-A19 harmonic parent charge), tabulated against external sequestering. **(4) Framing precision:** the old-CC problem is "addressed at the radiative-stability face, conditional," not "closed"; the Horizon-Landauer continuation is explicitly *not executed* and demoted to Appendix D with a numerology warning (gate F-F46.6). Verification 33/33 PASS + 11/11 guards (up from 14/14 + 6/6; Block AV action variation + Block CS status separation added), fail-closed.

Re-audited against The Book v11.0; no upstream value or status moved. Barrier B3: radiative-stability face addressed (conditional), residual-value face OPEN (B3-B); $\Omega_{\Lambda,0} = 83/121$ and U_N untouched.

v1.0 (July 2026): Initial public release. Executes the ZS-A26 entry-condition for the deferred vacuum-energy normalizer (N_{center}). Four results: T1 (the sequestering core — constant-shift cancellation in the $T - \langle T \rangle$ source, radiative stability; IMPORTED-PROVEN mechanism, DERIVED cancellation); T2 (the seam global constraint — one harmonic parent charge on the connected 38-node ZS-A19 seam graph, $\dim \ker L_\Gamma = 1$, offset-sensitive; DERIVED-CONDITIONAL on the ZS-A19 branching conditions); T3 (the finite-register trace normalizer — the ZS-F23 $\frac{1}{2} \ln 2$ template, coincident with the ZS-F19.6 seam handshake; DERIVED-CONDITIONAL on ZS-F23 Condition C); and T4 (the honest reduction — the radiative-stability face of the old CC problem is addressed, the residual-value face reduces to a single calibrated boundary datum = B3-B; NON-CLAIM on the absolute value). Five pre-registered gates F-F46.1–F-F46.5. Verification 33/33 PASS + 11/11 guards, fail-closed; 3 firewalled observations. Zero fitted parameters; (**A**, **Q**, dim **Z**) = (35/437, 11, 2) LOCKED. Barrier B3: radiative-stability face addressed, residual-value face OPEN (B3-B); $\Omega_{\Lambda,0} = 83/121$ and U_N untouched (breakthrough C). (Consolidated from internal Z-Spin Collaboration deep-exploration notes following ZS-A26 v2.2, ZS-F45 v1.3, and The Book of Z-Spin Cosmology v11.0.)